

HAMZA ASIM

AI Engineer | LLM Systems | Agentic AI | RAG and Document Intelligence

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PROFESSIONAL SUMMARY

AI Engineer building and deploying LLM applications, multi-agent systems, MCP-based tools, RAG pipelines, AI voice agents, document intelligence systems and workflow automation. Published researcher in multi-agent LLM systems and hybrid SLM-LLM architectures, with hackathon-winning AI projects. Strong background in Python, FastAPI, LangGraph, vector databases, REST APIs, microservices, Docker, Kubernetes and cloud deployment on AWS and GCP.

TECHNICAL SKILLS

LLM and Agents: RAG, GraphRAG, Multi-Agent Systems, Model Context Protocol (MCP), Tool and Function Calling, LangChain, LangGraph, LlamaIndex, LangSmith, Guardrails, LLM Evaluation, Reinforcement Learning

Retrieval and Knowledge: Embeddings, Vector Databases, Qdrant, FAISS, Chroma, Hybrid Search, Reranking, Knowledge Graphs, Neo4j

Document AI and NLP: OCR, AWS Textract, PaddleOCR, Tesseract, Layout Parsing, Table Extraction, Information Extraction, Text Classification, spaCy, Hugging Face Transformers, LoRA and PEFT Fine Tuning

Backend and Data: Python, FastAPI, REST APIs, Microservices, PostgreSQL, Redis, SQL, Snowflake, Apache Airflow, ETL Pipelines

Cloud and MLOps: Docker, Kubernetes, AWS (Bedrock, Lambda, S3, RDS), GCP, GitHub Actions, CI/CD, Twilio, Deepgram

PROFESSIONAL EXPERIENCE

AI Engineer, Docwyn.ai (Wyoming, USA, Remote)

Feb 2024 to Present

- Built and deployed an end to end OCR pipeline for an enterprise client with 20,000+ employees, turning daily documents into a searchable database.
- Raised extraction accuracy from 78% to 96% by pairing layout aware OCR with validation rules and confidence thresholds routing low certainty pages to review.
- Cut turnaround on daily document batches from 6 hours to under 45 minutes by parallelising extraction across worker queues and batching inference.
- Shipped an internal RAG assistant over the processed document store, letting operations staff answer record level questions in seconds.
- Orchestrated Apache Airflow pipelines consolidating data from 120+ US retail locations into Snowflake, cutting retrieval time 40% by rewriting query patterns and standardising reusable Python and SQL components.

Machine Learning Engineer, Omdena (New York, USA, Remote, Contract)

Jan 2025 to Jun 2025

- Built an agentic mental health support system on a four stage LangGraph pipeline (Triage, Retrieval, Response, Safety Validation), with conditional edges, retry limits and deterministic routing keeping clinical decisions out of the model's hands.
- Grounded replies in a RAG layer over curated mental health resources using sentence embeddings with FAISS and Chroma, replacing open ended generation with retrieval backed answers.
- Implemented structured risk scoring with Pydantic schemas across self harm language, hopelessness, severe distress and repeated high risk intent, escalating at a configurable threshold or immediately on critical cues.
- Routed escalations through FastAPI, PostgreSQL, Redis and Twilio/SendGrid to alert therapists with risk category, conversation ID and context summary, and traced agent paths, token use, latency and escalation events in LangSmith.

AI Engineer Intern, Glowing Soft (Lahore, Pakistan)

Jul 2023 to Oct 2023

- Built Python NLP pipelines processing 500K+ records per day, automating ingestion, cleaning and transformation and cutting manual work 40%.
- Applied classification and extraction to contract data, turning free form text into structured fields and saving 3 hours of review daily.
- Delivered 3 production ready models and tuned SQL workloads on AWS RDS by 35%, speeding up experimentation and client facing outputs.
- Cut downstream data defects 30% with automated schema and validation checks that caught malformed records before reporting.
- Cut model retraining turnaround from 2 days to 6 hours by packaging preprocessing and evaluation into a reusable workflow across 3 projects.

PROJECTS

Autonomous Voice Sales and Service Agent (CRM Integrated) | LLMs, FastAPI, Twilio, Deepgram, ElevenLabs, PostgreSQL, Docker, Kubernetes, AWS

- Built a multi-tenant voice agent for outbound sales and inbound support, pipelining telephony, speech to text, LLM tool calling and synthesis to hold turn latency near 800ms across 50+ concurrent calls.
- Connected it to CRM and product databases for live inventory, quotes and confirmed orders mid call, with transcript analysis covering intent, sentiment, churn risk and upsell signals written back to CRM.
- Trained a reinforcement learning layer over call outcomes to tune pitch sequencing and objection handling, lifting test conversion by 22%, and packaged the system as configurable microservices on Kubernetes so a new business onboards in under a day.

Multi-Agent Automation Platform with MCP Tooling | Python, MCP, FastAPI, LangGraph, Qdrant, PostgreSQL, Redis, Celery, Docker

- Designed a four agent system (Planner, Retrieval, Execution, Validation) that decomposes a natural language request into a typed task graph, routes each step to the right tool and verifies output before returning it.
- Exposed databases, vector search, internal REST services and file stores as MCP tools behind versioned schemas, so adding a capability means writing a server contract rather than editing agent logic.
- Raised multi step task completion from 62% to 89% with a validation agent that re-runs failed steps against schema checks, backed by exponential backoff retries and persistent state in PostgreSQL and Redis for resumable jobs.
- Cut token cost per workflow by 35% by routing simple lookups to a smaller model, caching retrieval results and reserving the larger model for planning, with step level trace logging for debugging.

PUBLICATIONS

The Agreeable Swarm: Sycophantic Drift, Consensus Cascades, and Epistemic Collapse in Multi-Agent Large Language Model Systems (2026).

Introduces the ESRD and MA-ESRD drift framework and the Sycophancy Amplification Factor (SAF), a metric for how far a multi agent swarm shifts from a correct position under peer pressure, plus mitigations such as adversarial reviewer roles and confidence weighted aggregation. [\[DOI\]](#)

When Small Models Create Big Overhead: The Specialization-Coordination Trade-off in Hybrid SLM-LLM Agent Systems (2026). Shows when splitting work across small specialist models stops paying off because coordination overhead outgrows the compute saved, and builds a break even framework across token, latency and handoff costs into architecture selection guidance for SLM, LLM and hybrid systems.

AWARDS

Google Gemini Hackathon (2024). Ranked in the global top 50 from 57,000+ participants across 160+ countries. Built an AI agriculture platform using NASA Earth-observation data, geospatial analytics and vision-language models for crop, yield and wildfire-risk insights.

EDUCATION

Bachelor of Science in Data Science, University of Central Punjab, Lahore

2022-2026

VOLUNTEER EXPERIENCE

Volunteer, Akhuwat Foundation. One month clothing donation drive; collected and organized donations for families in need.